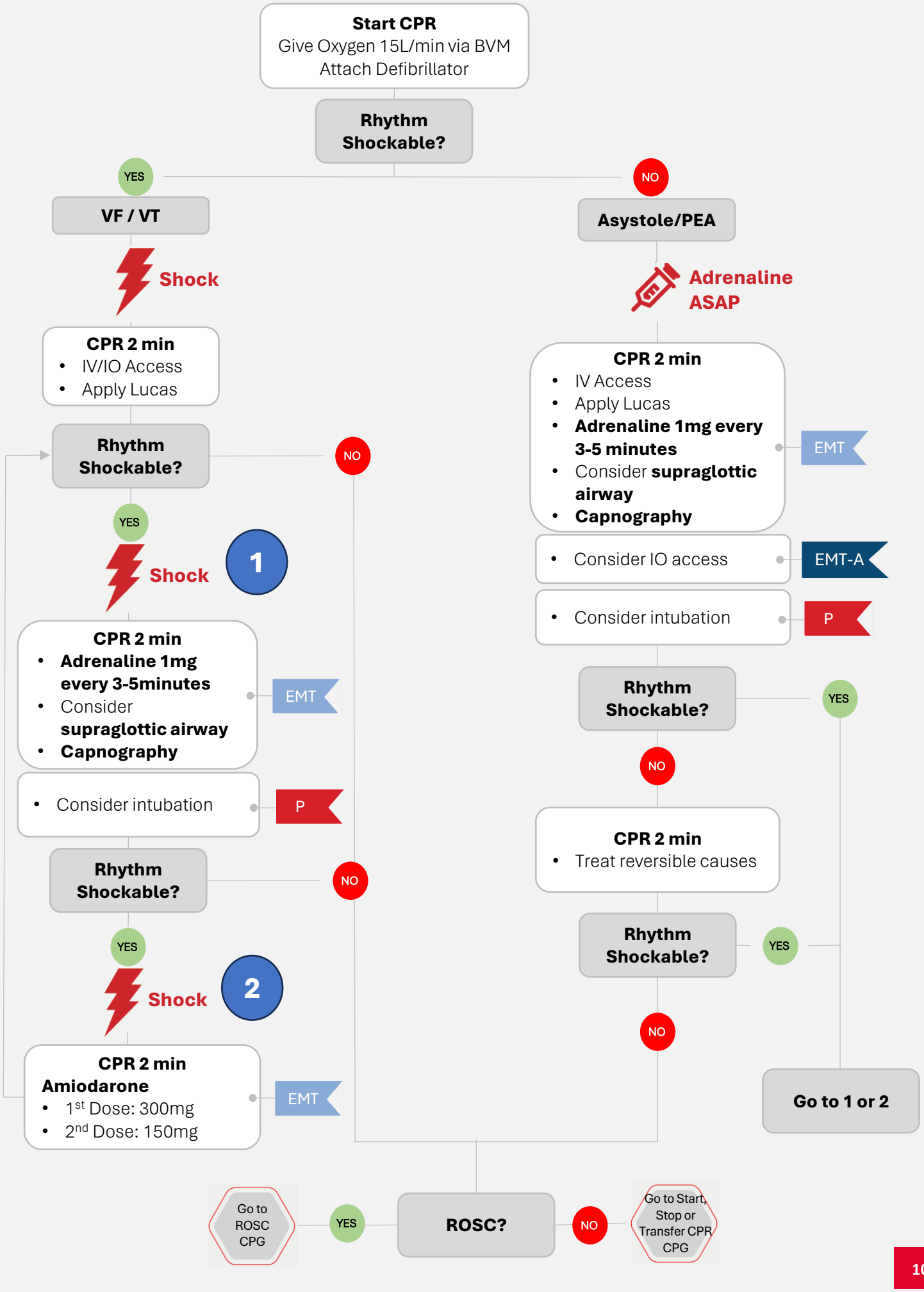
Adult Cardiac Arrest



Patient Care Goals

1. Rapidly identify and manage adult cardiac arrest to restore spontaneous circulation and oxygenation through high-quality cardiopulmonary resuscitation (CPR) and advanced life support (ALS) interventions.
2. Address reversible causes of cardiac arrest to improve survival and neurological outcomes.
3. Ensure seamless coordination with hospital care for post-arrest management.

Inclusion & Exclusion Criteria

Inclusion Criteria

- Adult patients (≥13 years) presenting with cardiac arrest, including:
    - Unresponsiveness with no signs of life (e.g., no pulse, no breathing, or agonal gasps).
    - Electrocardiogram (ECG) or automated external defibrillator (AED) rhythms indicating ventricular fibrillation (VF), ventricular tachycardia (VT), pulseless electrical activity (PEA), or asystole.
- Patients requiring immediate CPR, defibrillation, and ALS interventions to restore return of spontaneous circulation (ROSC).

Exclusion Criteria

- Patients with obvious signs of irreversible death (e.g., rigor mortis, dependent lividity, or decapitation).
- Pediatric patients (<13 years; refer to Pediatric Cardiac Arrest CPG).

Key Patient Assessment Points

1. **Cardiac Arrest Confirmation**
    - Verify unresponsiveness by assessing for no response to verbal or painful stimuli (AVPU).
    - Check for absence of pulse (carotid or femoral) for 5-10 seconds.
    - Simultaneously, confirm absence of normal breathing or presence of agonal gasps for 5-10 seconds.
2. **Rhythm Assessment**
    - Apply monitor/defibrillator to identify shockable rhythms (VF/VT) versus non-shockable rhythms (PEA/asystole)
    - Reassess rhythm every 2 minutes during CPR cycles to guide defibrillation or medication administration.
3. **Reversible Cause Identification**
    - Assess for reversible causes (H's and T's: Hypovolemia, Hypoxia, Hydrogen ion [acidosis], Hypo-/Hyperkalemia, Hypothermia, Tension pneumothorax, Tamponade, Toxins, Thrombosis [pulmonary/coronary]).
    - Obtain bystander history for potential causes (e.g., drug overdose, trauma, or witnessed collapse).
4. **ROSC and Post-Arrest Monitoring**
    - Check for signs of ROSC (e.g., spontaneous pulse, normal breathing, or rising end-tidal CO2 [EtCO2] >40 mmHg) every 2 minutes.
    - Monitor for re-arrest or deterioration post-ROSC to guide further interventions.
5. **Determine Triage Category**
    - Determine Triage Category as per Patient Triage Categories CPG.
        - All patients cardiac arrest are Red Priority.

Patient Management & Interventions

1. **Primary Survey**
    - Perform primary Survey as per Universal Care CPG.
        - Adapt for Cardiac Arrest: **Follow CAB:** Circulation, Airway, Breathing.
    - Assign roles for team-based resuscitation (e.g., compressor, ventilator, defibrillator operator, medication administrator).
    - Call for ALS back-up.
2. **Consider ALS back-up**
    - Consider the requirement for ALS back-up as per Universal Care CPG.
3. **Cardiac Arrest Management**
    - **Chest Compressions**

        - Initiate high-quality compressions (100-120/min, 5-6 cm depth, full recoil) immediately.
        - Rotate compressors every 2 minutes to prevent fatigue.
        - Minimize interruptions (<10 seconds for rhythm checks or defibrillation).
        - Mechanical chest compression devices
            - May be used when available.
            - Do not delay defibrillation to apply chest compression device.
    - **Airway Management**
        - OPA/NPA airway's should be used.
        - Placement of a Supraglottic airway is preferred over the use of an OPA/NPA if a supraglottic airway is available.
        - Patients being transferred under CPR should have a Supraglottic Airway in place wherever possible.
        - Intubation may be conducted in cardiac arrest provided that:
            - The team is properly briefed to minimize any interruptions in compressions and ventilation
            - Pre-intubation airway assessment predicts likely success on the first attempt.
        - Intubation may be conducted on predicted difficult airways if:
            - The Supraglottic Airway is not functioning well
            - Difficulties are being experienced using other airway adjuncts
            - There is vomit, blood or excessive secretions in the airway.
    - **Ventilation**
        - OPA/NPA in place:
            - Provide ventilations via bag-valve-mask (BVM) at a rate of 30 compression to 2 ventilations.
        - Supraglottic Airway/ Endotracheal Tube in place:
            - Provide uninterrupted ventilations via bag-valve-mask (BVM) at 10-12 breaths/min (1 breath every 5-6 seconds).
            - Attach end tidal carbon dioxide monitor (EtC02)
        - Avoid hyperventilation to prevent increased intrathoracic pressure.
    - **Oxygen**
        - Administer 100% oxygen via BVM or advanced airway to optimize oxygenation.
    - **IV Access**
        - Indicated for medication administration or fluid resuscitation during CPR.
        - Consider intraosseous (IO) access if IV access cannot be obtained
    - **Defibrillation**
        - Deliver first shock for VF/VT as soon as defibrillator is ready.
        - Deliver subsequent shocks for VF/VT once every 2 minutes.
        - Resume CPR immediately after shock (<10 seconds interruption).
    - **Waveform Capnography**
        - Use EtCO2 to monitor CPR quality (target >10 mmHg) and detect ROSC (>40 mmHg).
4. **Medication Administration**
    - **Adrenaline**
        - Adults: 1 mg IV/IO, every 3-5 minutes.
    - **Amiodarone**
        - 1st Dose: Adults: 300 mg IV/IO after the 3rd shock
        - 2nd Dose: Adults: 150 mg IV/IO after the 5th shock.
5. **Secondary Survey**
    - Assess for reversible causes during CPR cycles using the "5H's and 5T's" pneumonic:
        - Hypovolemia
        - Hypoxia
        - Hydrogen ion (acidosis)
        - Hypo-/hyperkalemia
        - Hypothermia
        - Tension pneumothorax

        - Tamponade, cardiac
        - Toxins
        - Thrombosis, pulmonary
        - Obtain history from bystanders (e.g., time of collapse, medical history, or precipitating events).
6. **Consider Red Flags**
    - Persistent shockable rhythm (VF/VT) after multiple defibrillation attempts and antiarrhythmics.
    - Suspected irreversible causes (e.g., massive myocardial infarction, end-stage disease).
    - Unsafe scene conditions (e.g., hazardous environment) preventing effective resuscitation.
    - Need for advanced procedures (e.g., ECMO, cath lab) requiring rapid transport.
7. **Patient Reassessment**
    - Reassess for ROSC
        - Reassess rhythm, pulse, and EtCO2 every 2 minutes to guide interventions and detect ROSC.
        - Transition to post-ROSC care if spontaneous circulation returns.
8. **Evaluate for Signs of ROSC**
    - Check for ROSC (e.g., spontaneous pulse, normal breathing, EtCO2 >40)

Patient Safety Considerations

- CPR Quality
    - Minimize interruptions (<10 seconds for rhythm checks, shocks, or ventilations).
    - Ensure compression depth (5-6 cm), rate (100-120/min), and full recoil.
    - Use EtCO2 or feedback devices to optimize CPR quality.
- Defibrillation Safety
    - Ensure no one touches the patient during shock delivery to prevent injury.
    - Remove patient from wet surfaces or conductive materials before defibrillation.

Clinical insights & Additional Guidance

- **Use of Manual Mode**
    - Advanced Paramedics and Emergency Physicians may elect to use the defibrillator in cardiac arrest in manual mode to minimize interruptions to chest compressions.
    - Significant risk exists of missing rhythm checks with using manual mode in cardiac arrest - a person must be assigned to keep track of rhythm check timing.
    - It is recommended that manual mode be used after key activities (such as airway and IV access) have been completed, and sufficient manpower is available to keep track of rhythm checks.
- **Shock Energy for Defibrillation**
    - If using manual mode:
        - Lifepak 15: 200 J, 300 J, 360 J (all further shocks at 360 J)
        - Zoll X-Series: 120 J, 150 J, 200 J (all further shocks at 200 J)
- **Ventilator Use**
    - It is appropriate to use a ventilator in cardiac arrest if within scope of practice.
    - See the Invasive Ventilation CPG for further information.
- **Resuscitation in Pregnancy**
    - Resuscitation in Pregnancy follows the same pattern as normal adult resuscitation with the following adaptations:
        - Position the patient Left Lateral or manually displace the uterus (to relieve aortocaval compression)
        - Perform chest compressions higher on the sternum (to account for the elevated diaphragm and heart displacement)
    - Peri-Mortem Cesarean delivery:
        - Patients with >20weeks gestation should be rapidly transported to a hospital capable of peri-mortem Cesarean Delivery (coordinate with Dispatch 2).
- **Hypothermia and Cardiac Arrest**
    - Withhold epinephrine until core temperature is >30°C.

    - If medications are to be given (e.g. in prolonged arrest with partial rewarming), use standard doses but extend intervals (e.g., every 6–10 minutes) to avoid accumulation and post-rewarming catecholamine surge.
    - Defibrillation doses remain unchanged.
    - All cardiac arrest patients secondary to hypothermia must be transferred to hospital under CPR.
- **The decision to transfer under CPR**
    - See Starting, Stopping or Transferring CPR CPG.

Documentation

- Identification: Patient's name, age, gender.
- Mechanism / Medical Complaint: Description of cardiac arrest event, witnessed or unwitnessed, bystander CPR status.
- Injuries / Information: History from bystanders, possible reversible causes (H's and T's), medical alerts.
- Signs: Initial rhythm, vitals, EtCO2 readings, ROSC indicators, neurological status if achieved.
- Treatment: Time CPR initiated, AED applied, defibrillation administered, ventilation method and rate, use of airway adjuncts, suctioning, oxygen, compressor rotation, IV/IO access, medications (e.g., adrenaline, amiodarone—document dose in mg).
- Allergies: Any known medication allergies.
- Medications: Recent medications or known cardiac history.
- Background: Medical history including known cardiac or respiratory conditions.
- Other: Time of arrest, downtime, scene factors, hospital prenotification, transport decision rationale.

Key Performance Indicators

1. Initiate chest compressions within 1 minute of recognition of cardiac arrest
2. Deliver the first shock within 1 minute of VT/VF rhythm recognition.

References

1. American Heart Association 2020, *2020 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care*, American Heart Association, Dallas.
2. Jarvis, J.L., Panchal, A.R., Lyng, J.W., Bosson, N., Donofrio-Odmann, J., Braude, D.A., Browne, L.R., Arinder, M., Bolleter, S., Gross, T., Levy, M., Lindbeck, G., Maloney, L.M., Matera, C.J., Wang, C.-T., Crowe, R.P., Gage, B., Lang, E.S. & Sholl, J.M. 2025, 'Evidence-based guideline for prehospital airway management', *Prehospital Emergency Care*, Taylor & Francis, viewed at https://www.tandfonline.com/loi/ipec20.